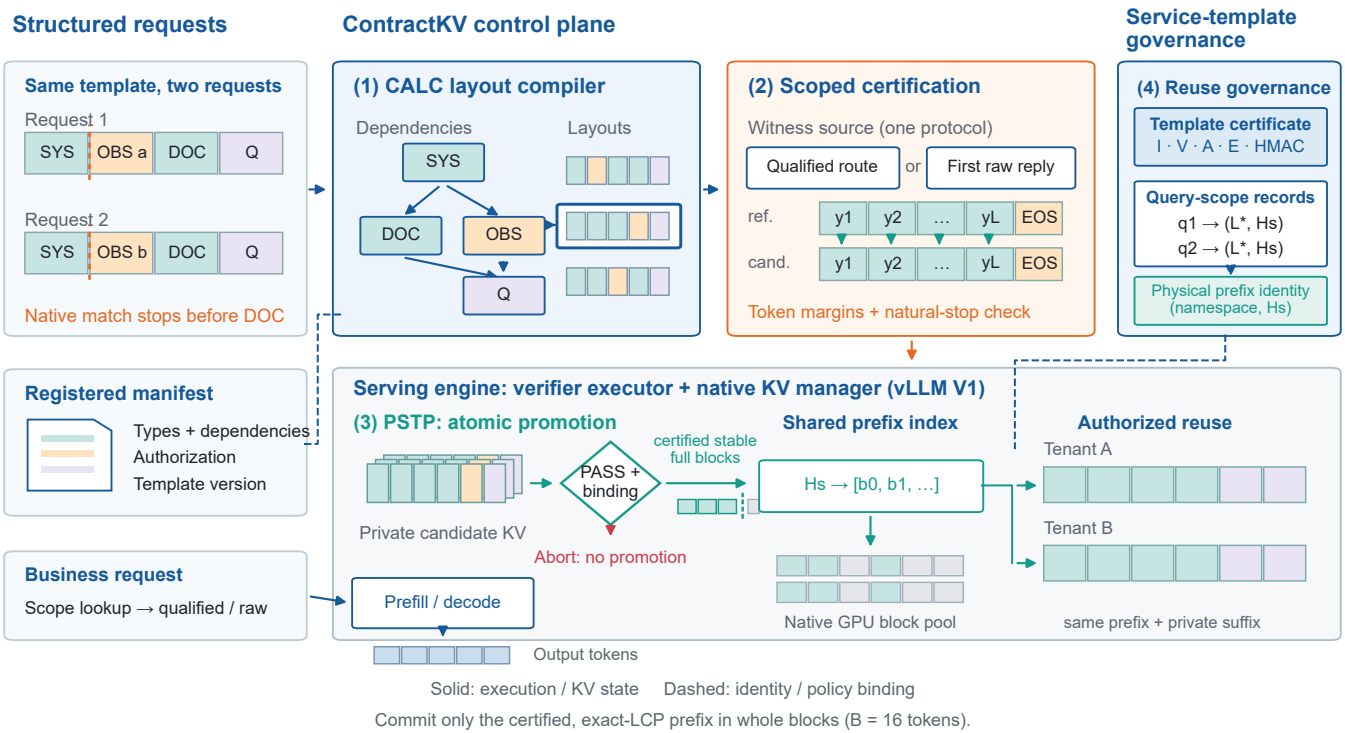

Structured requests
ContractKV control plane
Service-template governance
Same template, two requests
Request 1
SYS
OBS a
DOC
Q
Request 2
SYS
OBS b
DOC
Q
Native match stops before DOC
(1) CALC layout compiler
Dependencies
Layouts
SYS
DOC
OBS
Q
(2) Scoped certification
Witness source (one protocol)
Qualified route
or
First raw reply
ref.
y1
y2
…
yL
EOS
cand.
y1
y2
…
yL
EOS
Token margins + natural-stop check
(4) Reuse governance
Template certificate
I · V · A · E · HMAC
Query-scope records
q1 → (L*, Hs)
q2 → (L*, Hs)
Physical prefix identity
(namespace, Hs)
Registered manifest
Types + dependencies
Authorization
Template version
Serving engine: verifier executor + native KV manager (vLLM V1)
(3) PSTP: atomic promotion
Shared prefix index
Authorized reuse
Tenant A
Tenant B
Private candidate KV
PASS + binding
certified stable full blocks
Hs → [b0, b1, …]
Abort: no promotion
Native GPU block pool
same prefix + private suffix
Business request
Scope lookup → qualified / raw
Prefill / decode
Output tokens
Solid: execution / KV state
Dashed: identity / policy binding
Commit only the certified, exact-LCP prefix in whole blocks (B = 16 tokens).